

MICHAEL NDILOLO

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Aspiring chemical engineer at Hamilton College in a 3-2 dual-degree program culminating in a B.S. in Chemical Engineering from Columbia University. Developer of OlivAir, an independent industrial decarbonization platform with a filed U.S. provisional patent application. Gates Scholar and QuestBridge Scholar with research interests in industrial decarbonization, carbon removal technology, and applied chemistry.

EDUCATION

Hamilton College — B.A. Chemistry (3-2 Program with Columbia University) *Aug 2025 – May 2030 (expected)*

- Enrolled in Hamilton's 3-2 dual-degree engineering program: three years of a chemistry major at Hamilton followed by two years at Columbia University, culminating in a B.S. in Chemical Engineering

Miami Southridge Senior High School — High School Diploma

2023 – 2025

RESEARCH & INDEPENDENT PROJECTS

OlivAir — Independent Industrial Decarbonization Platform

2023 – Present

- Designing a reactor platform that uses industrial acid waste as the thermodynamic driver for mineral dissolution and CO₂ capture, producing permanent solid carbonate in place of neutralized sludge; documented across a ten-part technical paper series (TRL 2, analytically modeled)
- Filed U.S. Provisional Patent Application No. 64/124,248 (Aug. 2026), "Electrochemical Reactor with Isolated-Electrolyte Protected Electrode Chamber, Integrated Multi-Function Baffle Wall, and Dual-Mode Hydrodynamic Classification Zone," covering three platform mechanisms
- Built a full parametric CAD model of the reactor (STEP/STL solid-model export) from the platform's own design parameters, verified with an automated geometry completeness check
- In early technical dialogue with researchers at Yale University on the platform's electrochemical and mineralization design
- Originated as a passive, self-powered olivine-weathering device for carbon mineralization; placed 2nd as a team project at the Frost Science STEM Challenge (climate change track, Phillip and Patricia Frost Museum of Science)

EXPERIENCE

Recreation Equipment Hub Staff — Hamilton College

Aug 2026 – Present

- Manage and issue recreational and athletic equipment for the Hamilton College community

Student-Athlete, NCAA — Hamilton College, Clinton County, NY

Aug 2025 – Present

- Competed as goalkeeper for Hamilton College soccer (Year 1); transitioned to wide receiver and kicker for Hamilton College football (Year 2)

Hospice Care Intern — VITAS Healthcare, Miami, FL

Sept 2024 – March 2025

- Supported healthcare professionals in providing patient-centered hospice care to patients and families

HONORS & AWARDS

- Gates Scholar — Hamilton College
- QuestBridge Scholar — Hamilton College
- Posse Miami Finalist — Davidson College
- Frost Science STEM Challenge — 2nd Place (team project, climate change track)

CERTIFICATIONS

- Stanford Online: Thermodynamics and Phase Equilibria
- CPR / AED / First Aid — American Red Cross

SKILLS

Chemical Engineering · Hazardous Waste Management · Marine Conservation · Reactor & Process Design · Technical Writing